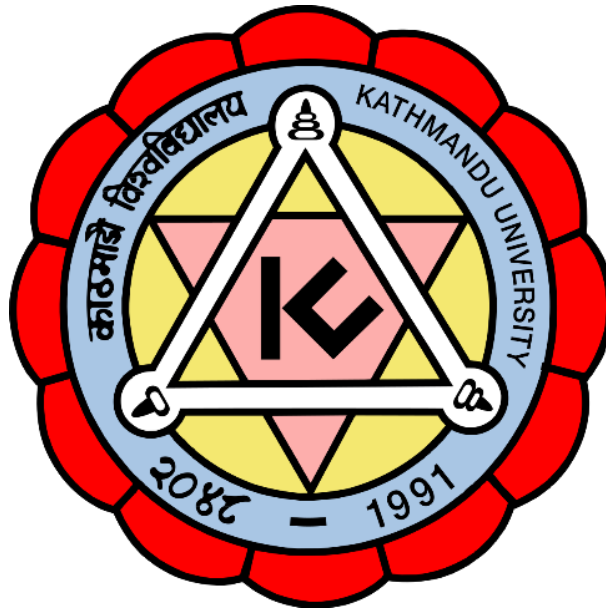


**KATHMANDU UNIVERSITY**  
**SCHOOL OF ENGINEERING**  
**DEPARTMENT OF GEOMATICS ENGINEERING**  
**DHULIKHEL, KAVRE, NEPAL**



**ENGINEERING PROJECT REPORT ON**  
**PHYSICAL PROTOTYPE MODEL OF KU CAFE**

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# ABSTRACT

Physical 3D models played an important role in engineering communication by transforming measured geometries into tangible representations that could be easily interpreted without the need for specialized software or digital visualization tools. This project involved the design and fabrication of a presentation-ready 1:25 scale physical model of the KU Cafe, located within the central premises of Kathmandu University, Dhulikhel, Nepal. The objective was to produce an accurate and visually appealing prototype that represented both the building and its immediate surroundings for academic presentation and exhibition purposes.

The project began with detailed site visits, during which field observations, photographs and direct measurements were collected. These data were processed into scaled drawings and fabrication layouts before being converted into a 1:25 scale model. The prototype was constructed on an 80 cm × 120 cm base using brown board and Styrofoam as the primary structural materials. Additional materials, including adhesives, printed textures and miniature decorative elements, were incorporated to improve the model's structural quality and visual appearance.

The surrounding environment, including pathways, landscaped areas, roads, seating spaces and other site features, was also represented to provide a comprehensive understanding of the cafe's spatial context and circulation. The completed model accurately reflected the building's architectural form, roof geometry and surrounding landscape while maintaining the selected 1:25 scale and an appropriate level of detail. The project enhanced practical skills in field measurement, scale conversion, engineering visualization, structural planning, model fabrication and teamwork. The final prototype was successfully completed within an eight week schedule at a total cost of NPR 6,500 and served as an effective tool for academic demonstrations, presentations and exhibitions.

**Keywords:** *physical prototype, scale model, ku cafe, geomatics engineering*

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# TABLE OF CONTENTS

|                                                        |     |
|--------------------------------------------------------|-----|
| Abstract .....                                         | i   |
| Acknowledgement .....                                  | ii  |
| Table of Contents .....                                | iii |
| List of Table .....                                    | v   |
| List of Figures.....                                   | vi  |
| List of Acronyms .....                                 | vii |
| 1 Introduction.....                                    | 1   |
| 1.1 Background.....                                    | 1   |
| 1.2 Problem Statement .....                            | 1   |
| 1.3 Objectives.....                                    | 1   |
| 1.3.1 Main objective.....                              | 1   |
| 1.3.2 Specific objectives .....                        | 2   |
| 1.4 Scope of Work .....                                | 2   |
| 1.5 Limitations.....                                   | 2   |
| 2 Literature Review .....                              | 3   |
| 3 Project Overview .....                               | 4   |
| 3.1 Study Design .....                                 | 4   |
| 3.2 Study Area .....                                   | 4   |
| 3.3 Data Source .....                                  | 5   |
| 3.3.1 Primary Data objective.....                      | 5   |
| 3.3.2 Secondary Data .....                             | 5   |
| 4 Methodology and Execution .....                      | 6   |
| 4.1 Site Observation and Photography .....             | 6   |
| 4.2 Dimension Measurement and Scaling .....            | 7   |
| 4.3 Sketch Preparation and Layout Planning .....       | 7   |
| 4.4 Material Selection and Component Fabrication ..... | 8   |
| 4.5 Structural Assembly and Detailing .....            | 8   |
| 4.6 Final Finishing and Presentation .....             | 8   |
| 5 Materials and Tools Used.....                        | 9   |
| 5.1 Tool Used.....                                     | 9   |
| 5.2 Materials Used .....                               | 9   |
| 6 Project Timeline and Detailed Budget .....           | 10  |
| 6.1 Project Timeline .....                             | 10  |
| 6.2 Budget and Cost Analysis .....                     | 10  |

|     |                              |    |
|-----|------------------------------|----|
| 7   | Results and Discussion ..... | 11 |
| 7.1 | Key Results .....            | 11 |
| 7.2 | Discussion.....              | 13 |
| 8   | Conclusion .....             | 14 |
| 9   | Recommendations .....        | 15 |
|     | References.....              | 16 |

## LIST OF TABLE

|                                     |    |
|-------------------------------------|----|
| Table 1: Gantt Chart .....          | 10 |
| Table 2: Detailed budget plan ..... | 10 |

## LIST OF FIGURES

|                                                                      |    |
|----------------------------------------------------------------------|----|
| Figure 1: Study Area Map.....                                        | 4  |
| Figure 2: Sectional view of Cafe .....                               | 5  |
| Figure 3: Workflow diagram.....                                      | 6  |
| Figure 4: Site Observation of KU Cafe .....                          | 6  |
| Figure 5: Base Level plan .....                                      | 7  |
| Figure 6: Digital 3D CAD Visualization in SketchUp.....              | 8  |
| Figure 7:Different angle view of KU Cafe .....                       | 12 |
| Figure 8: Final Constructed Physical Scale 3D Model of KU Cafe ..... | 13 |

## **LIST OF ACRONYMS**

|     |                       |
|-----|-----------------------|
| CAD | Computer-Aided Design |
| KU  | Kathmandu University  |
| NPR | Nepalese Rupee        |
| LED | Light Emitting Diode  |
| VR  | Virtual Reality       |
| 2D  | Two-Dimensional       |
| 3D  | Three-Dimensional     |

# 1 INTRODUCTION

## 1.1 Background

Modelling is the 'language of designing' (Roberts et al., n.d.) Physical modeling has been an effective method for communicating engineering, architectural and geomatics concepts by transforming measured spaces into tangible representations. Unlike digital models, physical prototypes provided a clearer understanding of scale, proportions, spatial relationships and the surrounding environment through direct observation.

The KU Cafe at Kathmandu University was selected as the study area because of its architectural significance and prominent location on campus. The project involved field measurements, photography and site observations, which were converted into scaled drawings before fabrication. A 1:25 scale physical model of the KU Cafe and its surrounding features, including pathways, landscaped areas, roads and seating spaces, was constructed on an 80 cm × 120 cm base using brownboard, Styrofoam and other model-making materials.

The completed model accurately represented the building and its surroundings while maintaining the selected 1:25 scale and demonstrating practical skills in engineering visualization, scale modeling, teamwork and prototype fabrication. It served as an effective tool for academic presentations and exhibitions.

## 1.2 Problem Statement

Interpreting multi-angled building geometry and the surrounding terrain through two-dimensional drawings or static digital visuals alone often led to spatial misunderstanding. Key architectural features such as roof geometry, circulation pathways and the relationship between the building and its immediate landscape were difficult to communicate effectively using these methods. In contrast, a physical model provided a clearer and more intuitive understanding of the structure and its surrounding environment. This highlighted the need for an accurate, durable and visually clear physical prototype that could be produced within limited time and budget constraints.

The project addressed the following challenges:

- ❖ It demonstrated that an accurate and detailed physical model could be developed within a limited budget and project duration by applying appropriate scaling methods, material selection and fabrication techniques.
- ❖ It improved the understanding of architectural drawings by bridging the gap between two-dimensional plans and a three-dimensional physical representation, allowing viewers to better visualize the building's form, proportions and spatial relationships.

## 1.3 Objectives

### 1.3.1 Main objective

- ❖ To design and construct a realistic physical scale 3D model of the KU Cafe.

### 1.3.2 Specific objectives

- ❖ To observe the KU Cafe and document its architectural and surrounding landscape features through field observations, photography and measurements.
- ❖ To collect structural dimensions and convert them into an appropriate scale for fabrication on a 120 cm × 80 cm base.
- ❖ To prepare scaled sketches, layout plans and a 3D SketchUp model to support accurate visualization and fabrication.
- ❖ To fabricate a detailed physical model of the KU Cafe using brownboard, Styrofoam, adhesives and other finishing materials.
- ❖ To represent surrounding features such as pathways, roads, landscaped areas, trees and seating spaces to enhance realism and spatial understanding.
- ❖ To apply engineering visualization, scale modeling and practical model making techniques throughout the project.
- ❖ To strengthen teamwork, planning, time management and project management skills during the design and fabrication process.

### 1.4 Scope of Work

- ❖ The project covered the exterior form of the KU Cafe and its immediate surroundings within the model base.
- ❖ The prototype included the building footprint, perimeter form, roof geometry and major openings to accurately represent the architectural characteristics of the structure.
- ❖ Surrounding site features, including pathways, roads, landscaped areas, trees and seating spaces, were incorporated to enhance realism and improve spatial understanding.
- ❖ The completed physical model was developed as an educational and presentation tool for academic demonstrations and exhibitions rather than as a detailed construction blueprint.

### 1.5 Limitations

- ❖ **Scale generalization:** Very fine architectural details were simplified due to the limitations of manual cutting and fabrication at a small scale.
- ❖ **Road levelling:** The actual road elevation partially obscured the front view of the building. Therefore, the road level was adjusted in the model to improve the visibility and overall presentation of the KU Cafe.
- ❖ **Material availability:** The selection of construction materials was limited to locally available and cost-effective options, which influenced the level of detailing and finishing.
- ❖ **Time constraint:** The project was completed within an 8-week academic schedule, limiting the opportunity for multiple design iterations, refinements and additional detailing.
- ❖ **Budget constraint:** The available budget restricted the use of premium materials and advanced fabrication techniques, requiring economical alternatives without compromising the overall quality of the model.

## 2 LITERATURE REVIEW

According to **(Roberts et al., n.d.)** modeling is regarded as the "language of design" because it transforms engineering and architectural concepts into tangible representations. Physical modeling remains an important communication tool in engineering, architecture and geomatics, as it enables measured spatial data to be visualized without relying solely on specialized software. Compared with two-dimensional drawings and digital renderings, physical models improve the understanding of scale, proportions, roof geometry and the relationship between a structure and its surrounding environment.

**(Fukuda et al., 2016)** highlighted that physical and digital modeling approaches are most effective when integrated. Computer-aided design (CAD) tools provide accurate scaling, layout preparation and dimensional verification, while physical fabrication converts verified digital designs into realistic prototypes. This combined workflow reduces interpretation errors and improves communication among engineers, designers, students and stakeholders.

According to **(Petrasova et al., 2015)**, the quality of a physical model depends greatly on material selection and finishing techniques. Lightweight materials such as plywood, foam board, Styrofoam are widely used because they are easy to fabricate, lightweight, and sufficiently durable. Proper surface finishing and the inclusion of surrounding elements such as pathways, vegetation and landscaping further enhance the realism and presentation quality of the model.

Similarly, **(Audrey Bardwell, 2024)** emphasized that physical models continue to play an important role in engineering and architectural education by bridging the gap between theoretical drawings and real-world structures. Project-based model making develops practical skills in engineering visualization, scale conversion, fabrication techniques, teamwork and project management while providing an effective medium for presentations and design communication.

### 3 PROJECT OVERVIEW

The project followed a practical engineering design approach involving site observation, field measurement, planning, scale conversion and physical model fabrication.

#### 3.1 Study Design

The project was conducted as a prototype-based engineering model development study. A scaled miniature model of the KU Cafe was developed using actual structural dimensions and surrounding environmental features obtained through field observations and measurements. The collected data were converted into scaled drawings and fabrication layouts, ensuring that the final model accurately represented the architectural characteristics and site context of the actual structure.

#### 3.2 Study Area

The study was conducted at the KU Cafe, located within the main campus of Kathmandu University, Dhulikhel, Kavrepalanchok, Nepal, approximately 30 km east of Kathmandu. The site was selected because of its architectural significance and accessibility for field observation and measurement.

The study area had the following characteristics:

- ❖ Geographic Coordinates: 27.618457° N, 85.538303° E
- ❖ Total Ground Area: 80.97 m<sup>2</sup>
- ❖ Perimeter: 40.60 m

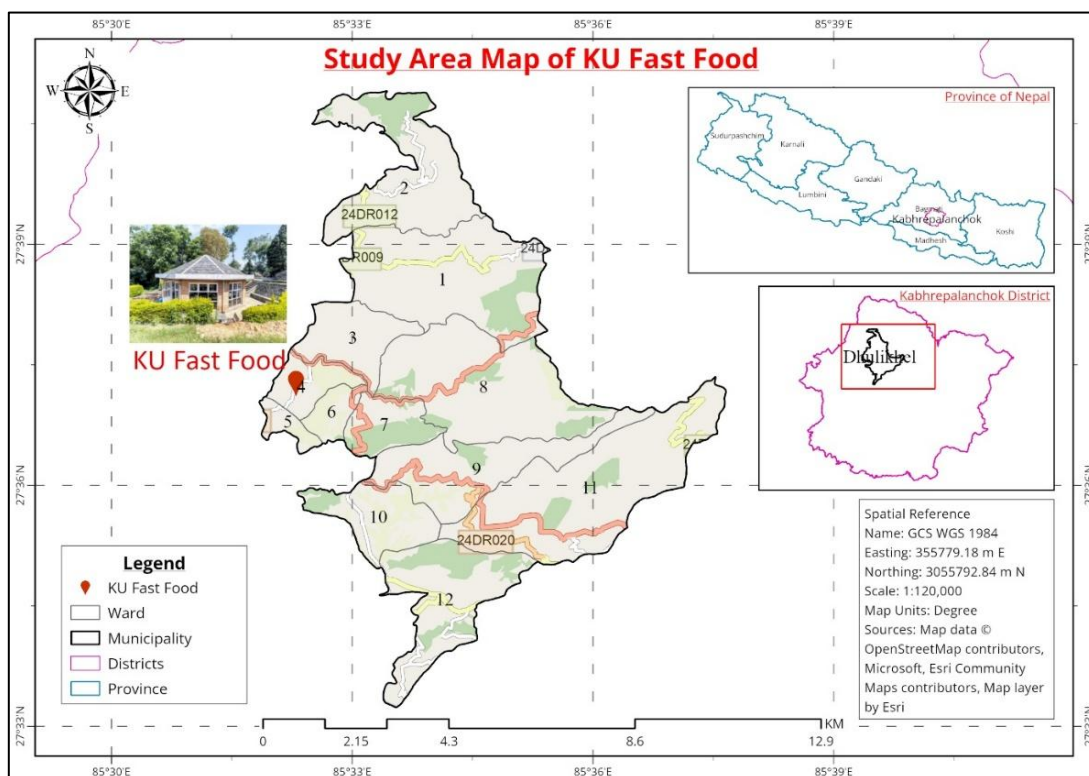


Figure 1: Study Area Map

### 3.3 Data Source

#### 3.3.1 Primary Data objective

- ❖ Ground-truth measurements taken on site with a measuring tape
- ❖ Walls, windows, doors and tables measured directly at KU Cafe
- ❖ Formed the base geometry converted to 1:25 scale

#### 3.3.2 Secondary Data

- ❖ Existing CAD drawing of KU Cafe obtained from Kathmandu University in PDF format

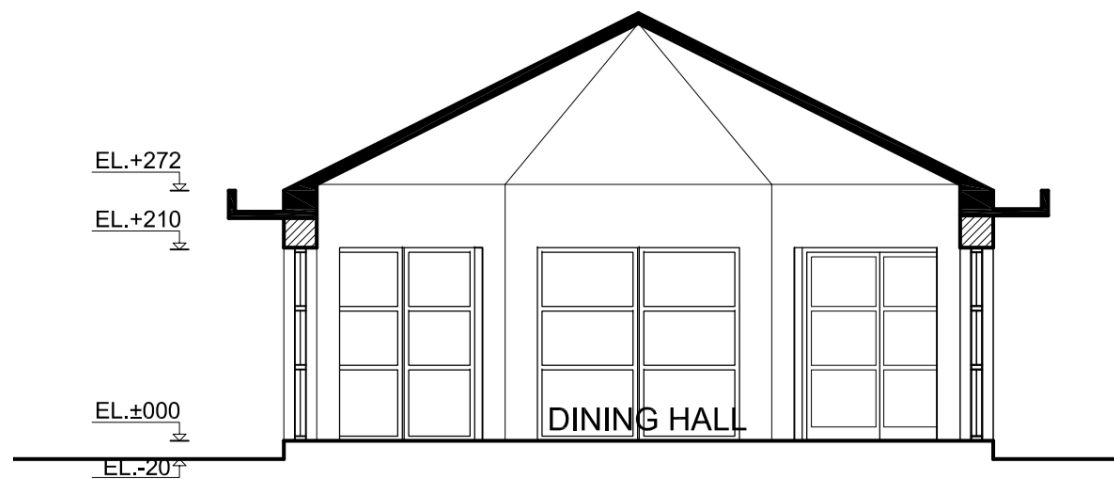


Figure 2: Sectional view of Cafe

## 4 METHODOLOGY AND EXECUTION

The project followed a systematic engineering workflow from site observation and measurement to model fabrication and final presentation. Each stage was carried out sequentially to ensure consistency between field measurements, scale conversion, material preparation and construction while minimizing dimensional errors.

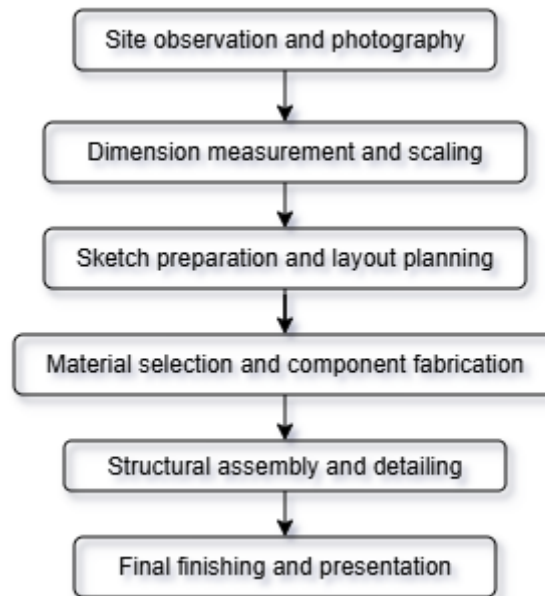


Figure 3: Workflow diagram

### 4.1 Site Observation and Photography

Multiple site visits were conducted to observe the KU Cafe and its surrounding environment. Photographs were captured from different viewpoints to document the building façade, roof geometry, pathways, landscaped areas and other important architectural features. These observations provided the necessary information for material selection, texture identification, color matching and the placement of surrounding site elements.



Figure 4: Site Observation of KU Cafe

## 4.2 Dimension Measurement and Scaling

Accurate field measurements were collected using measuring tapes. Key dimensions, including wall lengths, windows, door, table length and breadth were recorded and converted to a 1:25 scale for construction on an 80 cm × 120 cm model base. This scaling ensured that the proportions of the KU Cafe and its surrounding features were accurately maintained while fitting within the available base dimensions. The scaled measurements were repeatedly verified to minimize cumulative errors and ensure precision during the cutting, assembly and fabrication processes.

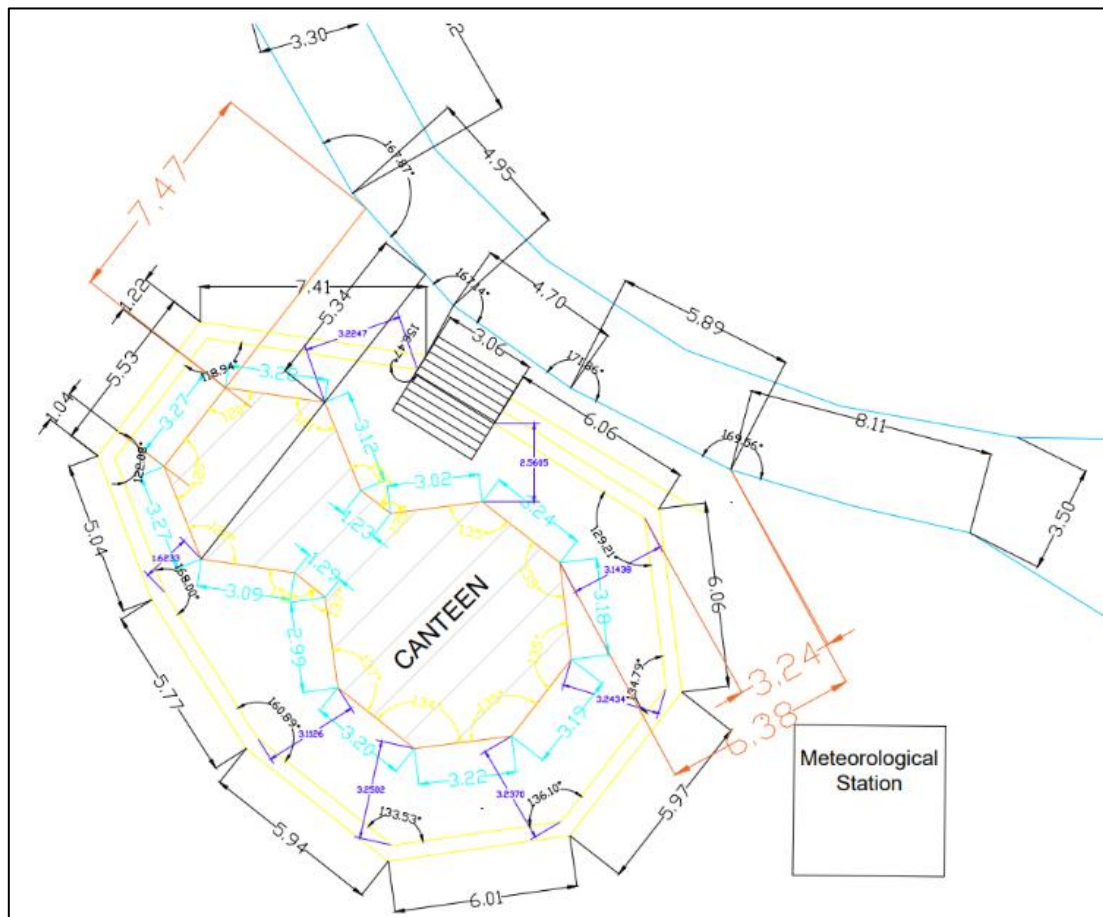


Figure 5: Base Level plan

### 4.3 Sketch Preparation and Layout Planning

Preliminary sketches scaled drawings and layout plans were prepared using the collected field measurements. A 3D SketchUp model was also developed to verify the building geometry before fabrication. These drawings established the building footprint, roof layout, pathways and surrounding landscape features, serving as references throughout the construction process.



*Figure 6: Digital 3D CAD Visualization in SketchUp*

#### **4.4 Material Selection and Component Fabrication**

Materials were selected based on their availability, affordability, ease of fabrication and finishing quality. Brownboard was primarily used for roof construction, while Styrofoam was used for wall sections and structural components. Additional materials, including adhesives, printed textures, artificial grass, acrylic sheets, paints and miniature decorative elements, were incorporated to improve realism and presentation quality. Individual components were manually cut, shaped and prepared before assembly.

#### **4.5 Structural Assembly and Detailing**

The model was assembled systematically by first fixing the building footprint onto the base, followed by the installation of wall sections and roof components. Adhesives were carefully applied to maintain structural stability and clean joints. Architectural details such as doors, windows and roof elements were represented according to the selected scale. Pathways, roads, landscaped areas, trees and seating spaces were then added to accurately represent the surrounding environment.

#### **4.6 Final Finishing and Presentation**

The finishing stage included applying printed wall textures, painting exposed surfaces, installing artificial grass and placing miniature trees and landscaping elements to enhance realism. Final adjustments were made to improve visual balance, cleanliness and presentation quality. The completed physical model accurately represented the KU Cafe and its immediate surroundings and was prepared for academic presentation and exhibition.

## **5 MATERIALS AND TOOLS USED**

Materials and tools were selected to balance accuracy, durability, cost and ease of fabrication under an 8-week academic schedule.

### **5.1 Tool Used**

- ❖ Measuring tape and scale ruler
- ❖ Cutter
- ❖ Glue gun
- ❖ LED lighting components
- ❖ Drawing and drafting tools

### **5.2 Materials Used**

- ❖ Plyboard
- ❖ Styrofoam, Brown Board
- ❖ Decorative ground-texture materials
- ❖ Fiber Glass
- ❖ Electric bulb, wire

## 6 PROJECT TIMELINE AND DETAILED BUDGET

The project was completed over 8 weeks. Tasks were organized to progress from planning and measurement to fabrication, finishing and presentation preparation.

### 6.1 Project Timeline

Table 1: Gantt Chart

| Activities                      | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---------------------------------|---|---|---|---|---|---|---|---|---|----|
| Literature Review & Planning    |   |   |   |   |   |   |   |   |   |    |
| Site Visit & Field Survey       |   |   |   |   |   |   |   |   |   |    |
| 2D CAD & 3D Digital Layout      |   |   |   |   |   |   |   |   |   |    |
| Material Procurement            |   |   |   |   |   |   |   |   |   |    |
| Base Plate & Wall Fabrication   |   |   |   |   |   |   |   |   |   |    |
| Roof, Windows & LED Integration |   |   |   |   |   |   |   |   |   |    |
| Landscaping & Fine Details      |   |   |   |   |   |   |   |   |   |    |

#### Legends



Completed Task

### 6.2 Budget and Cost Analysis

Project expenditures were maintained within the allocated budget. The total budget was NPR 13,560, including VAT.

Table 2: Detailed budget plan

| SN                | Materials/Items              | Estimated Cost (NPR) |
|-------------------|------------------------------|----------------------|
| 1                 | Foam Board/Brown Board       | 1000                 |
| 2                 | Acrylic Sheet                | 1,500                |
| 3                 | Adhesives and Glue Gun       | 1000                 |
| 4                 | LED Lights                   | 1500                 |
| 5                 | Artificial Grass and Trees   | 500                  |
| 6                 | Stationery and Cutting Tools | 930                  |
| 7                 | Miscellaneous Expenses       | 500                  |
| Total cost (NRS)  |                              | 6830                 |
| Grand Total (NRS) |                              | 6830                 |

## 7 RESULTS AND DISCUSSION

The primary deliverable was a completed physical prototype of the KU Cafe constructed on a 80 x 120 cm base. The finished model was presentation-ready and communicated the building's overall massing, distinctive roof geometry and its relationship to nearby circulation and landscape elements.

### 7.1 Key Results

- ❖ **Roof geometry and form** — The roof planes and overall silhouette were fabricated to reflect the KU Cafe's characteristic multi-plane roof structure, supporting clear recognition of the building at model scale.
- ❖ **Pathways and circulation contexts** — were represented to indicate pedestrian movement and spatial connections around the structure.
- ❖ **Landscaped site context** — Artificial grass and miniature trees were incorporated to reflect the immediate environment and to improve readability of site boundaries and open space zones.

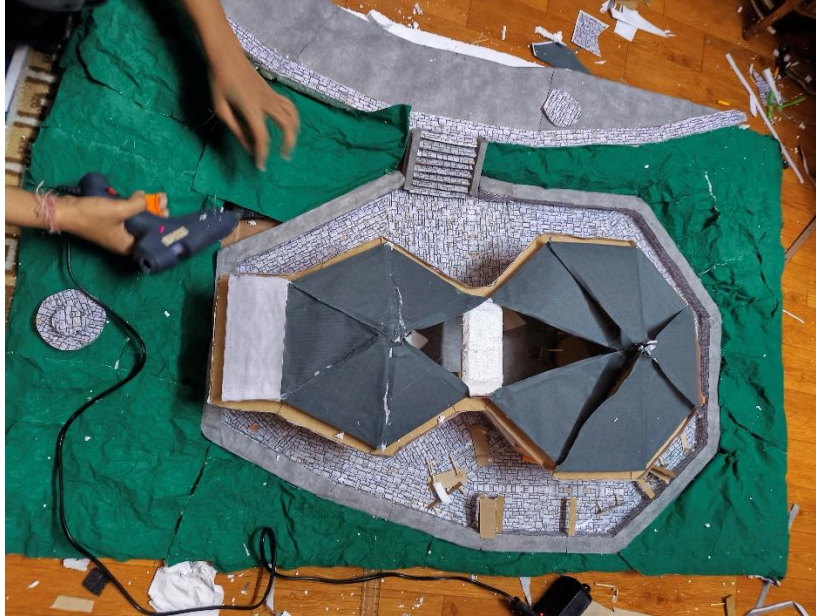




*Figure 7: Different angle view of KU Cafe*

## 7.2 Discussion

The fabrication process demonstrated that careful measurement transfer and consistent scaling were essential to maintaining perimeter accuracy and roof alignment. Manual cutting introduced minor edge irregularities, but these were reduced through trial fitting, controlled adhesive application and surface finishing. The use of brownboard and Styrofoam enabled lightweight construction while maintaining adequate stiffness for handling and display. Landscaping elements improved overall realism and helped situate the building within a recognizable campus like setting, consistent with the final model appearance observed in the completed prototype image.



*Figure 8: Final Constructed Physical Scale 3D Model of KU Cafe*

## **8 CONCLUSION**

This project successfully bridged theoretical spatial concepts with practical hands-on engineering execution. By converting raw survey measurements of the KU Cafe into a physical scale prototype, the project addressed common challenges in spatial visualization and material construction within budget and time constraints.

The final physical model provides a clear, presentation-ready 3D representation for the Kathmandu University community. It highlights how physical prototypes remain invaluable tools alongside modern digital drafting tools in engineering education.

## 9 RECOMMENDATIONS

For future teams seeking to extend the prototype's technical and presentation value, the following recommendations were proposed:

- ❖ **Improve precision fabrication** — Use laser cutting or 3D printing for repeated components and complex joints to improve edge quality and consistency at small scale.
- ❖ **Modularize the roof** — Design a detachable or hinged roof mechanism to allow optional viewing of simplified interior zoning while maintaining exterior integrity.
- ❖ **Enhance site detail** — Add more standardized ground textures and clearer pathway markings to improve legibility of circulation and terrain transitions.
- ❖ **Strengthen documentation** — Maintain a step-by-step fabrication log with measured checkpoints to make the model reproducible and to support assessment rigor.

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